SIT723 Research Techniques & Applications
Task 3.1P Research Questions

Overview

This research investigates whether large language models (LLMs) produce stable semantic interpretations when the same meaning is expressed through different but equivalent wording. The paper is dedicated to the field of clinical natural language processing, in which the reliability is particularly critical since fluctuating results can pose hazards in medical practice. A semantic entropy framework based on UMLS, proposed in the project, will be used to assess the consistency of input variations that are mapped by LLMs to the same semantic interpretation, meaning preservation. The research will integrate biomedical datasets, controlled perturbations, and ontology-based output clustering to go beyond the traditional accuracy-only evaluation and offer a more accurate measure of model behaviour.

Research Strategy

This research paper is guided by a systematic assessment plan that is informed by the themes and gaps in research that were outlined in the previous literature review assignments. The main dataset will be MedMentions ST21pv since it offers UMLS-linked biomedical concepts, whereas SQuAD 2.0 and BioASQ Task B Phase A (2021-2023) will serve as secondary validation datasets, allowing to conduct a domain-continuum analysis between clinical concept normalisation and biomedical QA and general-domain QA. The methodology is a combination of meaning-preserving perturbation generation, six-gate automated quality testing, semantic entropy quantification, concept mapping with UMLS, and in-scale comparisons between biomedical and general-purpose language models. The analytical priority of the research questions is arranged, and the questions are formulated in a way that will yield quantifiable results that can directly contribute to the thesis.

Research Questions

1. Which linguistic properties of meaning-preserving perturbations perturbation type, magnitude of lexical change, and grammatical category of modified tokens are the strongest independent predictors of semantic entropy in clinical LLM outputs, and do these predictors interact?

- How does semantic entropy vary across perturbation types (back-translation, controlled paraphrase, synonym substitution, and syntactic reordering) when all other linguistic factors are held constant?
- Does the magnitude of lexical change, measured as normalised Levenshtein edit distance, independently predict semantic entropy after controlling for perturbation type and grammatical category?
- Do different grammatical categories (nouns, verbs, modifiers, function words) contribute differently to semantic entropy, and does a perturbation-type by grammatical-category interaction exist?

This is the main research question in that it studies what is behind instability in clinical interpretation of LLM at the level of specific linguistic variables instead of semantic entropy being a single aggregate measure. Possible current semantic entropy research, such as Kuhn et al. (2023, ICLR) and Farquhar et al. (2024, Nature) have not broken-down entropy by language dimension and have not causally analyzed which input properties cause variation. Semantic entropy, as found in the literature review (Task 2.3D, Gap 2), has not yet been systematically anchored in biomedical concept systems like UMLS, and no research has viewed the systematically in a clinical language context the linguistic properties of meaning-preserving perturbations that contribute to instability. The gap is explicitly filled in this question by applying UMLS-grounded CUI clustering as the criterion of equivalence and a pre-registered mixed-effects regression $\hat{H}(x_0) \sim \text{PerturbationType} \times \text{LinguisticCategory} + \text{Magnitude} + \text{Accuracy} + (1|\text{Instance}) + (1|\text{Model})$ to determine the strongest independent predictors and test the interactions. The pre-determined minimum effect size is $f\,2 \geq 0.04$ of fixed effects of interest.

Expected Deliverables:

- Semantic entropy scores broken down by perturbation type, lexical change magnitude, and grammatical category of modified tokens
- Mixed-effects regression results showing independent effect sizes (f^2) for each linguistic predictor with 95% bootstrap confidence intervals
- Test of the perturbation-type by grammatical-category interaction term and its practical significance

- Identification of which linguistic factor most strongly drives semantic instability in clinical concept normalisation

2. Does semantic entropy detect systematic interpretation instability in high-accuracy LLM outputs that traditional evaluation metrics fail to reveal, and does this accuracy–stability dissociation hold across both encoder-only and generative model architectures?
    - Can a model achieve high task accuracy while still producing inconsistent semantic outputs across meaning-equivalent inputs specifically in the high-accuracy quartile of the output distribution?
    - Does the accuracy–stability dissociation appear in both encoder-only models (CUI-ranking entropy) and generative models (text-generation entropy), or is it architecture-specific?
    - How often do apparently correct outputs remain semantically unstable when measured through UMLS-grounded semantic entropy rather than surface accuracy alone?

This is the most clinically significant of the three questions as it specifically tests whether current benchmark evaluation is misleading in its risk of reliability. Models used in clinical NLP are chosen and relied on depending on criteria of accuracy. Nonetheless, as shown by Sclar et al. (2024) and Raj et al. (2023), high benchmark accuracy may co-exist with a large degree of semantic instability in case of meaning-preserving variation. As identified in the literature review (Task 2.3D, Gap 1), the existing systems of evaluation are overly dependent on benchmark accuracy despite the numerous studies demonstrating that accuracy may conceal instability when meaning-preserving variation occurs; and as identified in Gap 4, there is currently no ontology-grounded, structured measure of evaluating the illusion of understanding in clinical language contexts. This question does not just focus on one gap, but by testing whether a good-performing model on accuracy can also exhibit high normalised semantic entropy H 0 (x 0) in the high-accuracy quartile, with a pre-defined effect size threshold rank-biserial r 0.30. The cross-architecture scope condition guarantees that the finding is generalised to more than one model family.

Expected Deliverables:

- Comparison of task accuracy and normalised semantic entropy $\hat{H}(x_0)$ across the same outputs, per model

- Statistical evidence of whether high-accuracy instances show significantly elevated semantic entropy (Wilcoxon signed-rank, one-tailed, rank-biserial $r \geq 0.30$)
- Separate analysis for encoder-only (CUI-ranking entropy) and generative (text-generation entropy) architectures
- Results demonstrating whether semantic entropy provides reliability information that accuracy metrics cannot detect

3. Does domain-specific biomedical pretraining reduce semantic entropy relative to general-purpose models at equivalent parameter scale, and does this stability advantage generalise from medical concept normalisation to biomedical and general-domain question answering?

Domain Adaptation and Semantic Stability

- Do biomedical models show lower semantic entropy than general-purpose models of equivalent scale in within-scale comparisons: BioBERT vs. BERT-base (both 110M parameters) and BioMistral-7B vs. FLAN-T5-XXL (large-scale tier)?
- Does any stability advantage observed on MedMentions (medical concept normalisation) persist on BioASQ (biomedical QA) and attenuate on SQuAD 2.0 (general-domain QA), indicating domain-specific rather than fully general stability benefits?
- Does domain adaptation improve the consistency of semantic interpretation under meaning-preserving variation, or does it only improve benchmark accuracy?

The question is concerned with comparing models and selecting principled models to use in safety-critical clinical NLP applications. It is established in the literature of biomedical NLP that domain-adapted models like BioBERT (Lee et al., 2020, Bioinformatics), PubMedBERT (Gu et al., 2021, ACM TCHC), and BioMistral (Labrak et al., 2024, ACL) are more accurate in tasks compared to general-purpose models. Nonetheless, as discovered during the literature review (Task 2.3D, Gap 3), biomedical LLMs are not compared systematically in the context of semantic stability with controlled meaning-preserving perturbation, and it is unclear whether an advantage regarding semantic consistency is also present. The within-scale comparison design BioBERT vs. BERT-base and BioMistral-7B vs. FLAN-T5-XXL controls the impact of domain-specific pretraining by removing the confounding effect of model scale. The pre-specified generalisation condition testing the assumption that the stability advantage diminishes on SQuAD 2.0 is pre-specified as part of this research question and will be tested with one-tailed Mann-Whitney U tests with pre-specified effect size threshold rank-biserial r ≥ 0.30 in each within-scale pair. Weakening of the benefit on general-domain QA would suggest that domain adaptation only provides task-specific, not entirely general, stability benefits, which is also a theoretically significant and clinically practical result.

Expected Deliverables:

- Within-scale semantic entropy comparisons: BioBERT vs. BERT-base (110M) and BioMistral-7B vs. FLAN-T5-XXL (large-scale), with effect sizes and 95% bootstrap confidence intervals
- Semantic entropy results across all three datasets (MedMentions, BioASQ, SQuAD 2.0) enabling a domain-continuum analysis
- Statistical evidence of whether domain adaptation improves semantic stability as well as task accuracy
- Discussion of whether any stability advantage is domain-specific or generalises across task formats, with implications for model selection in clinical NLP

References

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